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Fifteen Years of Data and Information Quality Literature: Developing a Research Agenda for Accounting

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ABSTRACT: This paper provides a framework for guiding accounting-related Data and Information Quality (DIQ) research, based on four major research strands: *people and decision-making, governance, operations, and technology (PGOT)*. The last three have been broken down further into three subtopics each, a total of ten subcategories. With people connecting the four strands, the resulting PGOT framework provides a structure to create DIQ research questions. DIQ-related articles published between the years of 1994 and 2008 were identified, the predominant research focus and method were determined. The coding identified research areas that need further exploration. Traditionally, DIQ research has been pursued by non-accountants. Accounting-oriented DIQ literature tends to concentrate on the decision aspects of the PGOT. With an increased emphasis on compliance, CobiT, and internal controls the accounting discipline can make a substantial contribution to the DIQ field, particularly with respect to the decision-making context within the relevant environment.

Keywords: data and information quality (DIQ); accounting information systems (AIS); decision-making; framework; research methods; PGOT.

I. INTRODUCTION

As a fundamental transaction process system (TPS), the accounting information system (AIS) collects data that eventually evolves into information products (IPs), such as sales reports and financial statements. Data is frequently exported into data marts and data warehouses, paired with data from other internal and external sources, and summarized and pro-

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cessed into reports that include information such as trend and market analyses to further support management decision-making. Since stakeholders make decisions in different environments, this results in varying data and information quality requirements.

Investors and lawmakers require corporations to provide accurate, timely, and complete financial statements. Citizens are demanding greater accountability from their governments, health care systems, and non-profits. As a result, these organizations are feeling pressure to demonstrate greater measurable impact, resulting in Data and Information Quality (DIQ) issues becoming increasingly important. Unfortunately, poor quality data exists everywhere. Some data is simply incorrect, such as errors on invoices or within inventory records. Other data is irrelevant, incomplete, out-of-date, or isolated in data islands. Collectively, across all organizational contexts, DIQ costs businesses billions of dollars per year (Birman 2003). Much of this cost is due to the fact that DIQ is multidimensional (Wang and Strong 1996) and, from a decision-making perspective, can vary depending on the context in which it is used. DIQ may suffice in one decision-making context but the same data and information may be inadequate in another.

Gauging how aware stakeholders are in the data and information they use to support decision-making is very important. For much of the data generated by an organization, the fields of accounting and AIS have the ability to positively impact the quality of data and information on which stakeholders rely. Likewise, the field of DIQ is capable of making a positive impact on the field of accounting and the effectiveness of AIS.

In this paper we seek to classify and motivate research addressing the following essential question:

> **Question:** How can we plan, design, and implement policies, procedures, systems, and information products within an appropriate context, such that the quality of the data and information will most effectively support decision-making?

This paper reviews research related to DIQ and explores areas where the fields of accounting and AIS can substantially contribute. Specifically, this paper provides a detailed framework for guiding DIQ research; the taxonomy is based on four major research strands. Within the framework are areas where accounting and AIS-related DIQ research should be more focused. DIQ-related articles published between the years of 1994 and 2008 were identified and for each of the 193 articles the predominant research focus and method were determined. Furthermore, the researchers coded each article based on research question subtopics. The coding was used to identify research areas that require further exploration. The next section provides a research framework that decomposes DIQ into four major categories: *people and decision-making, governance, operations and technology (*PGOT*)*. Section III details the taxonomy of research related to DIQ and AIS, organized by the people, governance, operations, technological and decision-making aspects of the DIQ research framework. The paper concludes with a discussion of future research opportunities within the accounting and AIS domains, particularly with respect to under-represented areas within the PGOT framework.

II. RESEARCH FRAMEWORK

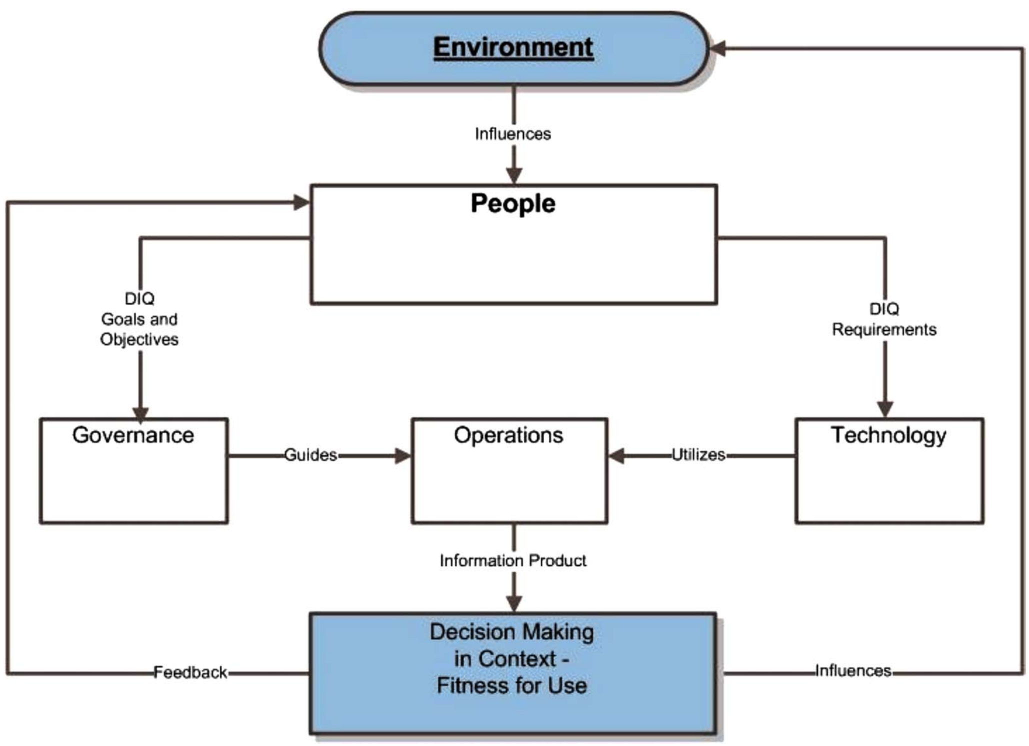
Surprisingly, although DIQ is of great importance to all organizations, no universal definition of what constitutes DIQ exists. Both researchers and practitioners agree that data quality is multidimensional and often impacted by the context in which the data is used (Klein 2000; Redman 1998; Wang and Strong 1996). What makes the DIQ field difficult to define is that DIQ touches all areas of an organization, much like accounting.

The framework described herein facilitates the classification of DIQ research in its broadest sense, examining those organizational components that impact the quality of data and information. Ultimately, data and information are only useful if it can be used by people within a context to improve decision-making, since it is the environment in which the decisions are made that ultimately impacts the perception of quality. The DIQ research framework described in the current paper is an adaptation derived from Wang and Strong (1996), Wang et al. (1995), Nord et al. (2005), and Poston and Grabski (2000). It consists of four general areas: *people and decision-making within context*, *governance*, *operational*, and *technology* (PGOT). See Figure 1 for a conceptual overview of the components and interactions within the PGOT framework.

As a result of the broad impact of DIQ on organizations, the existing DIQ literature draws upon multiple disciplines such as accounting, organizational behavior, business law, ethics, operations, and information technology, while simultaneously crossing many organizational boundaries. As a result, interdisciplinary research can lead to improvements in the DIQ field. The development of a DIQ research framework is motivated by a desire to categorize this existing DIQ literature and provide boundaries for analyzing DIQ literature with an accounting or AIS focus.

Given the interdisciplinary nature of DIQ, the authors drew on literature from the fields of accounting, management information systems (MIS), computer science, and software engineering in developing the PGOT framework, allowing researchers to examine DIQ through a variety of lenses that encourage interdisciplinary work. Although the framework is interdisciplinary, the focus of future research in this paper will be on the under-researched areas related specifically to accounting and AIS. Next, some of the relevant work that contributed to the development of the

FIGURE 1
DIQ in Context: PGOT Research Framework



PGOT is discussed. There will be a particular focus on the interrelationships between DIQ and the concept of information as a product, DIQ in relation to accounting and AIS, and DIQ and information technology management and control.

Relationship between DIQ and Treating Information as a Product

A review of the literature in any given field shows us both where we have been and where we need to go. A road map coupled with specific research questions provides structure to this process. Wang et al. (1995) reviewed the early data quality literature through 1995. The literature was categorized within a framework built upon research, which proposed that information should be treated as if it were a product. This intangible product takes raw materials (data) and transforms it through various processes to result in a product that can be used. This information product (IP) can take many forms, such as a sales report or purchase orders. The IP is used in the decision-making process within the context of the individual making the decision. The research by Wang et al. (1995) describes a framework for data quality comprised of seven key elements: management responsibilities, operation and assurance cost, research and development, production, distribution, personnel management, and legal function.[1] These elements were expanded upon in the general PGOT framework.

Relationship between DIQ and Fitness for Use

One of the primary goals of accounting and AIS is to provide useful, relevant, and timely information for both managerial and financial decision-making. Ultimately, judging the quality of data and information is dependent on how it will be used for decision-making (i.e., fitness-for-use). Given the heavy emphasis in accounting on decision-making, Joseph Juran's (1988) concept of fitness-for-use was incorporated into the PGOT framework (under decision-making). Ultimately, the data and information provided by an information system must be fit for use by those who utilize them to make decisions (i.e., consumers). Thus, the same data and information that are fit for one use can be inappropriate for another. Juran (1988) developed a short list of questions that organizations and individuals can use for determining whether a product or service are fit for use. Within the confines of this paper, the "product" under consideration is an information product (IP). Hence, in the context of DIQ, the questions for consideration are:

- *Who* uses the information product (IP) and who is responsible for its quality?
- *What* are the consumers' specific determinants of the fitness-for-use of the IP?
- *How* will consumers use the IP?
- What are the *economic resources* of collectors, custodians, and consumers?
- What possibility exists for endangering *human safety*?

Each of the four major PGOT areas will be examined in greater detail. Decision-making in context is threaded throughout the framework, examining each PGOT area in terms of Who, What, How, Economic Resources, and Human Safety. Research that is primarily focused on the decision-making component can be considered a fifth research strand for classification purposes. This framework is later used to code the existing literature into subcategories, allowing for a richer opportunity to define research questions and classify the existing research. After the literature has been coded, a more refined diagram of the PGOT is provided to summarize the detail discussed below.

[1] Other models of the scope of AIS research have been proposed by Murthy and Wiggins (1999), and Poston and Grabski (2000) among others. Some of the elements used in the PGOT arise from these models.

People

With respect to the DIQ research framework, the "P" in the *PGOT* stands for people. There are five roles people play: collectors, custodians, consumers, auditors, and managers. Each of these people will ultimately be a decision maker within some context. Collectors capture and enter data and information into the information system (IS). Custodians develop and maintain the systems that store the data and information needed to create information products (IPs). They are responsible for developing IPs that will meet the needs of consumers (end-users). Consumers make financial and managerial decisions using IPs. For a particular decision, the context will be at the individual, group, organizational, or global level. Auditors verify the data and information found in the systems as well as validate internal accounting controls. DIQ managers plan, organize, staff, lead (or direct), and control the resources needed to accomplish desired DIQ goals and objectives. Even though the roles above are described independently, people may fulfill many functions simultaneously.

Governance

Processes relevant to DIQ governance involve (1) plans, policies, and procedures, (2) legal and ethical compliance, and (3) personnel management. Plans, policies, and procedures are concerned with the creation, approval, and funding of corporate data and information quality plans, policies, procedures, and systems as well as upper management's role in the DIQ system. Legal and ethical focuses primarily on compliance with laws, regulations, and ethical codes of conduct. Personnel management involves making employees aware of the importance of DIQ through training, motivating them to comply with DIQ policies and procedures, measuring employees' DIQ achievement, and understanding the consequences (economic or otherwise) of noncompliance. Certain issues cut across all three areas. For example, data ownership must be determined when developing plans, polices, and procedures; there may be legal issues with respect to who owns the data; and it behooves the organization to incorporate data ownership into the training of personnel.

Operations

Operations are further refined into (1) production, (2) distribution, and (3) operations and assurance services. Given the regulatory requirements associated with accounting (e.g., GAAP, IFRS) and legislation, such as the Public Company Accounting Reform and Investor Protection Act of 2002 (i.e., Sarbanes-Oxley) and tax law, the auditing of the processes that transform data into information is a critical component. In addition, in some industries, such as health care and banking, privacy is heavily regulated making it a special concern along with the quality of the data. Production encompasses items such as the quality requirements necessary for the procurement of raw data, data structures, and records needed for the production of IPs. Production also ensures the process that transforms data into information is correct. Data should not change as a result of the production process. Lastly, production involves the utilization of data tags and extract-transform-load (ETL) tools as well as the identification of non-conforming data items and specifications of corrective action.

The movement of data and IPs through the system (i.e., distribution) is critical and includes such topics as data integration, metadata, quality documentation, and records management. As data is increasingly used in secondary sources (Wixom and Watson 2001), the need for contextual data tags and metadata has become vital to determining DIQ. Additionally, as the volume of data has exploded, the need for adequate records management processes has become increasingly crucial. As an example, consider how difficult it is to find a necessary document or file on a poorly managed flash drive. Multiply the volume of data stored in corporate databases by millions of records, and it is clear that this is an important concern.

Operations and assurance services pertains to the services provided by an independent party to validate or improve DIQ or its context. Operations and assurance services include prevention, appraisal, and failure costs related to the demonstration and proof of quality as required by customers and management. Furthermore, it includes such services as business risk assessment, information systems security review, customer satisfaction surveys, internal and continuous auditing, and accounts receivable review.

Technology

From a technology perspective, there are three components: input, systems and tool design (the "black box"), and output. The design of information systems should include input components that ensure data is not corrupted through the system process. Validation rules, data quality tools, and metadata tools all help to support this desired result. Systems from this perspective are independent of data and focus on the design of a superior system that produces quality data rather than on the data itself. With respect to the black box, systems should incorporate emerging technologies as they become economically feasible. Examples of such technologies include XML, XBRL, and ebXML. These tools will support the quality of the data by standardizing the movement process through the business system. The output component of the framework pertains to the analysis and design of the quality aspects of IPs (also referred to as semantics). From both an input and output perspective, issues related to DIQ dimensions, measurement/metrics, and fitness-for-use are all important.

Relationship between DIQ, Accounting, and AIS

DIQ and accounting are entwined. Although much of the early work in DIQ is found in MIS journals, there are numerous examples within the accounting literature reviewed that draw on this early work. Data tags, a common concept in the DIQ literature (Chengalur-Smith et al. 1999; Fisher et al. 2003; Neely 2002) are implemented in XBRL in the accounting realm (Bovee et al. 2002). Data audits in the DIQ field and continuous auditing in the accounting field are both concerned with ensuring that data is accurate (Alles et al. 2008; Bricker and Maydanchik 1999; Pierce 2004; Debreceny et al. 2005). Data champions are recognized in accounting (Tee et al. 2007) and MIS (Wixom and Watson 2001). Ultimately, the field of accounting is concerned with fairly representing the financial status of an organization, and the accuracy and reliability of the resulting financial statements allows stakeholders to make informed decisions.

Two critical components of any accounting information system are the quality of the data residing within the system as well as in the system's outputs. Accounting has a dramatic impact on the quality of data and information through the design and implementation of systems and processes. Accuracy is a particularly prevalent quality dimension when considering accounting data. However, data quality is actually multi-dimensional (Wang and Strong 1996). When considering DIQ with respect to AIS, it is important to consider the multiple dimensions of quality. Wang and Strong (1996)) identified 15 such dimensions (i.e., accuracy, believability, objectivity, reputation, value-added, relevancy, timeliness, completeness, appropriate amount of data, interpretability, ease of understanding, representational consistency, concise representation, accessibility, and access security). Other researchers have developed similar lists of dimensions. The fitness-for-use of the data and information should also be considered.

Given the multi-dimensionality of DIQ, it is important to manage the multi-attribute trade-offs within a cost-benefit relationship. Dimensions of accuracy (reliability), relevance, and understandability of the data and information are of critical importance in accounting. However, depending on the context, other dimensions may be highly valued as well. In some cases, timely information is imperative (e.g., demand data or stock prices); in another, investors require accurate financial statements. Dimensions often require trade-offs. For example a secure system, one that is on a corporate intranet, may not be accessible to potential customers or supply chain partners. These issues are not unique to AIS; context and fitness-for-use are concerns for all systems that generate information.

Relationship between DIQ and CobiT

The Control Objectives for Information and related Technology (CobiT) was created in 1996. CobiT is a set of best practices, originally for IT controls, but subsequently for the governance of information technology in general. CobiT focuses on streamlining business processes to improve efficiency and effectiveness of IT in an enterprise. CobiT has four focal points: (1) planning and organization; (2) acquisition and implementation; (3) delivery and support; and (4) monitoring and evaluation (Tuttle et al. 2007). CobiT 4.1 contains a core component called information criteria that is related to the dimensions of DIQ described above. The CobiT framework identifies the following seven information criteria: effectiveness, efficiency, confidentiality, integrity, availability, compliance, and reliability.

As described earlier, Wang and Strong (1996)) identified 15 dimensions of quality. These are grouped into four categories of quality important to end-users: intrinsic, representational, accessibility, and contextual (Kahn et al. 2002). Within these categories are the quality dimensions, all described in CobiT, albeit with alternate names. The definitions of the seven criteria of CobiT are essentially concerned with components of the PGOT. For example, effectiveness deals with information being relevant and pertinent to the business process as well as being delivered in a timely, correct, consistent, and usable manner. The definition of efficiency corresponds to the production and distribution of information products. As a last example, compliance correlates with the PGOT component of legal and ethical compliance. While the CobiT framework is well suited to its purposes, it does not lend itself well to the classification of the DIQ body of literature.

As the narrative above clearly illustrates, the PGOT framework, composed of multiple research strands, encapsulates the many issues and broad impact of DIQ on the organization. Identifying specific components helps to organize the existing research and examine patterns, thus helping to direct future research. In the next section the authors define terms, state research goals, describe the types of content analysis used, define the article selection criteria, describe the coding procedure, and discuss the inter-coder reliability.

III. TAXONOMY OF RESEARCH RELATED TO AIS AND DIQ

This study performed content analysis on published articles from within the DIQ research literature over the 15-year period from 1994 to 2008. The DIQ literature spans multiple disciplines including, but not limited to, MIS, accounting, management, computer science, software engineering, operations research/management science, and library science. A total of 193 articles were coded, recording research methods used and key findings.

Due to the quantity of articles reviewed, many resources associated with this paper are located at http://aaapubs.org/. Table 6, which is one of the online resources, contains definitions of the research methods used. Also online, a complete listing of the 193 articles with research methods and key findings can be found in Tables 7 through 16. Relevant papers are presented in tabular format arranged chronologically, by author within year. Each table represents a different research strand of the PGOT. In addition, 173 research questions were identified, tied to the individual components of the PGOT. This taxonomy can also be found in the online resource in Tables 17 through 19. In illustrating the richness of the PGOT framework, the authors refer to some of the research questions developed. The identification of specific research questions helps to place the existing research in context, while at the same time providing direction for future research within the field.

Content Analysis Methodology

The steps of content analysis (adapted from Gray and Densten 1998, Hair et al. 1998, Krippendorff 1980, Neuendorf 2002) include (1) fully describing the phenomenon being studied, in this case the research questions addressed and the primary research methods used within the DIQ literature; (2) selecting the level of analysis—DIQ research articles to be studied (see subsection entitled Article Selection); (3) deriving coding categories and operational definitions for research questions (see Table 1 online) and research methods coding categories and operational definitions; (4) deciding on a sampling strategy (see Article Selection for details); (5) training the coders (or raters) ensuring reliability; (6) gathering the data (counting existence or absence of some characteristic, forced choice, etc.); and (7) analyzing the data (see Section IV for results and recommendations for future research).

A major obstacle to any content analysis is determining the true or intended meaning of the document; in the case of this research, documents were peer-reviewed journal articles from the DIQ literature. In many cases, the authors did not explicitly list the research questions addressed within the articles, resulting in the need to use latent content analysis. However, in most cases researchers did explicitly state what research method or methods were used (i.e., manifest content analysis was used). In some cases, an author never stated what method was used and in a few instances, the author incorrectly classified the research method utilized. In these cases, the coders were required to infer the methods used (i.e., latent content analysis). Regardless of which of the two types of content analysis were used, meaning always has the potential to be misinterpreted. However, with three coders using the same operational definitions, as was the case in this study, the probability of this occurring is minimal.

Article Selection

In order to classify the DIQ literature, the Proquest-ABI/Inform, EBSCO databases, as well as the ACM Digital Library were searched for peer reviewed journal articles that contained the keywords "data quality" or "information quality." The results of this search consisted of "traditional" DIQ articles primarily from the MIS field. Furthermore, the coders reviewed 171 articles in the Journal of Information Systems (Fall 1995–Fall 2008) as well as 123 articles in the International Journal of Accounting Information Systems (Volume 1 Issue 1–Volume 10 Issue 2). These journals yielded additional articles that were not located with the initial keyword search, but are relevant with respect to DIQ. Issues such as effective data retrieval and query methodologies were deemed consistent with DIQ in accounting and AIS. Bibliographies from the papers were also examined, yielding additional papers for inclusion in this study. The search resulted in 193 articles being examined and coded for the current paper.

Coding Procedure

Three coders were used in this study, two of whom are authors of this paper. The two researchers and one research assistant read and coded each of the 193 articles selected from the DIQ literature, identifying research questions, key results, and primary research methods used. The steps taken in preparation of coding included:

- Defining research questions and operational definitions for each research method (see online Table 6 for research methods and Tables 17–19 for research questions http://aaapubs.org/)
- Classifying each research question into a subcategory within the PGOT framework
- Characterizing each method according to a well-defined research method classification
- Agreeing to a minimum acceptable level of reliability
- Training each coder in the use and interpretation of the coding instrument

The coders agreed to a 0.85 minimum acceptable level of reliability for research methods and 0.75 for the research questions. According to Landis and Koch (1977), a 0.81 to 1.00 represents almost perfect agreement. In fact, since there are so many research questions, the 0.75 is quite reasonable. The coders' training consisted of four sets of five articles each. After coding each set of five articles, the three coders discussed their assignments for each article looking for omissions or errors in the coding instrument and the corresponding definitions or for differences in how the coders interpreted or used the instrument. When disagreements arose, the team discussed and resolved them, making minor adjustments as needed. A few minor issues were identified and resolved. The Fleiss' kappa for the research methods was 0.8722, which is above the agreed to minimum level and the Fleiss' kappa for the research questions was above the agreed to threshold as well.

Coding

Each strand of the PGOT, other than people and decision-making, has three subcategories, thus allowing the coders to organize the research into ten subcategories in total. As shown in Figure 2, an expanded version of the PGOT, major DIQ components include: (1.1) plans, policies and procedures, (1.2) legal and ethical compliance, (1.3) personnel management, (2.1) the production and (2.2) distribution of information products, (2.3) operations and assurance services, (3.1) system input, (3.2) systems and tool design, (3.3) information products (i.e., output), and these components are managed by people in an environment that supports (4.0) decision-making in context. Further pairing of each of the ten sub-categories with dimensions that define Juran's (1988) concept of fitness-for-use (e.g., who, what, how, economic resources, and human safety) results in 50 possible research areas. Table 1 outlines these research strands and their corresponding subcategories. In Table 1, the column labeled "Who" corresponds to people. This structure allows for the enumeration of a rich collection of research questions (see Tables 17–19 of the online resources (http://aaapubs.org/) for a complete list of questions).

The PGOT framework logically displays the many inter-connected areas that pertain to DIQ. To summarize, *people* define DIQ goals and requirements, which are governed by *policies and procedures*, within *legal and ethical* parameters, and *personnel management* to ensure that everything is monitored by individuals who have been properly trained. In light of the goals and objectives, people define the DIQ requirements, which can be implemented through *technology*. *Technology* (*input-process-output*) is utilized to implement *operations* that, in turn, are guided by *governance*. *Operations*, via *production*, *distribution*, and a*ssurance services* will result in information products that can be used in decision-making within a given context. The decision-making is influenced by the environment and provides feedback to people for further improvement.

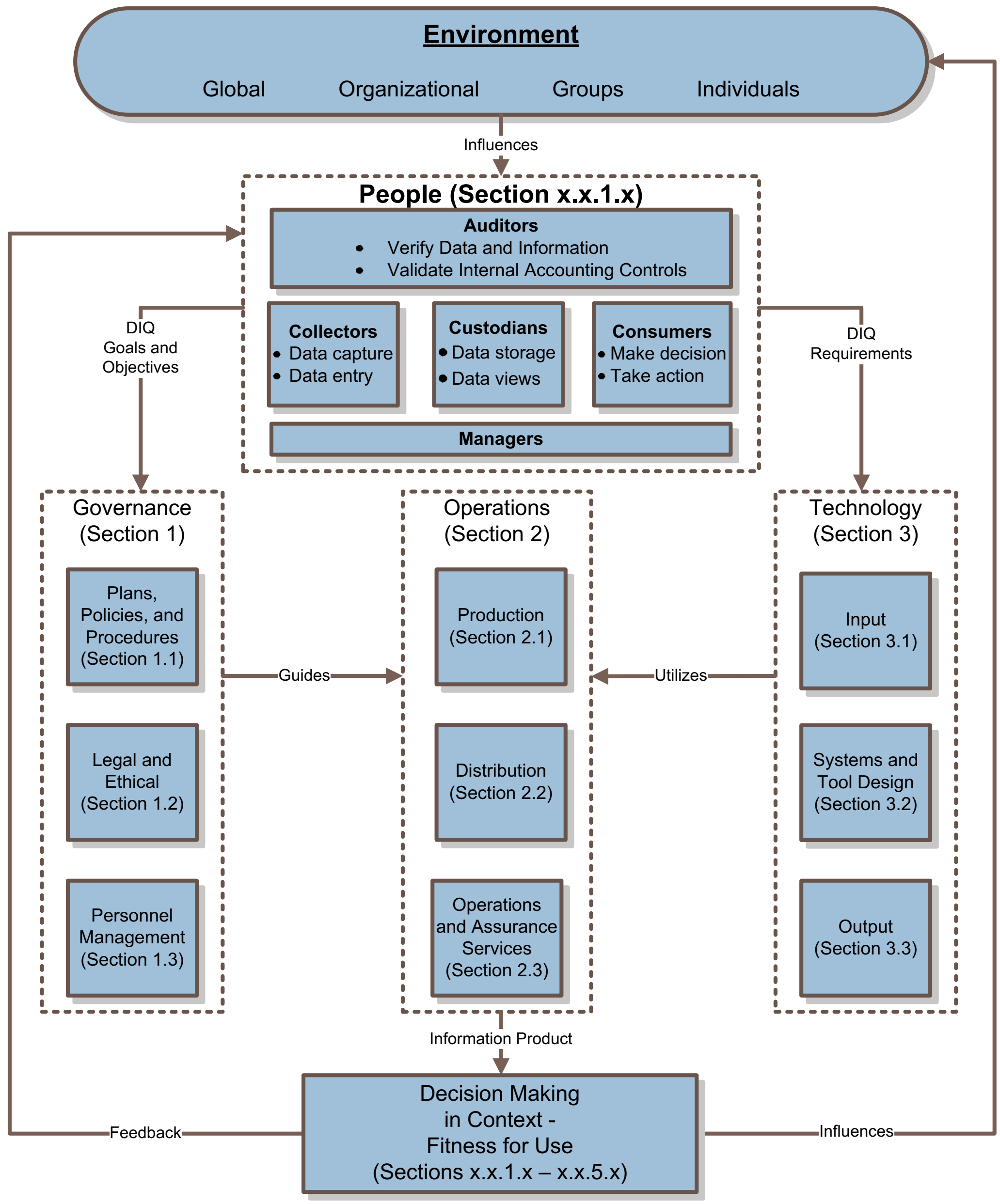
The research strands developed in the previous section were designed with a twofold purpose. The research strands provide a framework for categorizing 193 articles in the DIQ literature and, as indicated earlier, provides a foundation for a set of questions that can be used to code the articles in greater detail.

The next sections of the paper are organized as follows: each research strand within a section of the PGOT framework is described. An analysis of the questions indicates that the top 20 percent of questions is addressed in over 60 percent of the articles (see Table 2). Selected papers that address this top 20 percent of research questions are highlighted. Additionally, approximately 20 percent of the questions are not addressed in any of the reviewed articles. These questions represent opportunities for further research and will be described in Section IV.

Governance Strand—Plans, Policies, and Procedures (Section 1.1)

Research in this area deals with the creation, approval, and funding of corporate data and information quality plans, policies, procedures, and systems as well as upper management's role in

FIGURE 2
DIQ in Context: PGOT Research Framework



the DIQ system. There are several key roles that auditors and accountants can play in this realm. With their domain knowledge and corporate expertise, accountants and auditors have knowledge that should be tapped when forming teams to address both the creation and the implementation of

TABLE 1
Research Question Numbering Scheme for PGOT Taxonomy

Research Strands		Fitness-for-Use Categories				
		Who	**What**	**How**	**Economic Resources**	**Human Safety**
Categories	**Subcategories**	**x.x.1.x**	**x.x.2.x**	**x.x.3.x**	**x.x.4.x**	**x.x.5.x**
Governance	1.1 Plans, Policies and Procedures	1.1.1.x	1.1.2.x	1.1.3.x	1.1.4.x	1.1.5.x
	1.2 Legal and Ethical	1.2.1.x	1.2.2.x	1.2.3.x	1.2.4.x	1.2.5.x
	1.3 Personnel Management	1.3.1.x	1.3.2.x	1.3.3.x	1.3.4.x	1.3.5.x
Operations	2.1 Production	2.1.1.x	2.1.2.x	2.1.3.x	2.1.4.x	2.1.5.x
	2.2 Distribution	2.2.1.x	2.2.2.x	2.2.3.x	2.2.4.x	2.2.5.x
	2.3 Operations and Assurance Services	2.3.1.x	2.3.2.x	2.3.3.x	2.3.4.x	2.3.5.x
Technology	3.1 Input	3.1.1.x	3.1.2.x	3.1.3.x	3.1.4.x	3.1.5.x
	3.2 Systems and Tool Design	3.2.1.x	3.2.2.x	3.2.3.x	3.2.4.x	3.2.5.x
	3.3 Output	3.3.1.x	3.3.2.x	3.3.3.x	3.3.4.x	3.3.5.x
Decision-Making within Context (General Fitness-for-Use)		4.1.1.x	4.1.2.x	4.1.3.x	4.1.4.x	4.1.5.x

plans, policies, and procedures. Additionally, DIQ policies should be an explicit part of the development of AIS, with implementation and measurement a clear-cut result. In addition, procedures for automating the policies should become part of both audit programs and AIS.

Four of the top 20 percent of questions addressed in the existing literature appear in this research strand. They are as follows:

- "What plans, policies, and procedures should exist in order to ensure DIQ (1.1.2.1)?"
- "What is the noneconomic impact of DIQ on business, work processes, and strategy (1.1.2.2)?"
- "How are data quality systems integrated into the overall corporate structure and culture (1.1.3.3)?"
- "What are the financial, operational and legal risks associated with poor DIQ (1.1.4.2)?"

These questions have been addressed in multiple domains, including government, accounting, and health care. Divorski and Scheirer (2001) write in reaction to the Government Performance and Results Act (GPRA). They discuss how federal agencies are developing useful practices to verify and validate their performance information. There should be validation and verification, review of organizational capabilities, assignment of clear responsibilities, and adoption of data collection methods that encourage objectivity. Henderson and Murray (2005) focus on customer relationship management (CRM). They provide recommendations on prioritizing and delivering guidance on how to improve data quality as a crucial component of CRM. Tee et al. (2007) argue that DQ champions must change work processes and establish a data quality awareness culture in order to improve data quality. Redman (1998) speaks from experience when he states that "poor data quality increases operational cost because time and other resources are spent detecting and correcting errors. The cost incurred by the customer service organization to correct customer addresses, orders, and bills is a typical example." Terris and Litaker (2008) state, "Pay for performance programs and other initiatives directed at changing the behavior of health systems and

TABLE 2

Top 20 Percent Most Frequently Addressed Research Questions

Number	Question	Count
1.1.2.1	What plans, policies, and procedures should exist in order to ensure DIQ?	10
1.1.2.2	What is the noneconomic impact of DIQ on business, work processes, and strategy?	12
1.1.3.3	How are data quality systems integrated into the overall corporate structure and culture?	10
1.1.4.2	What are the financial, operational and legal risks associated with poor DIQ?	11
2.1.2.1	What production processes exist or are needed to ensure data is correct?	12
2.1.2.4	What interventions are available to improve the quality of data (e.g., training, managerial direction, incentives, outlier reports)?	19
2.1.3.1	How are the processes created and put in place to ensure correct data as output?	11
2.1.3.2	How is data cleaning achieved?	17
2.1.3.3	How do you implement data integrity?	15
2.1.3.4	How are data errors detected?	24
2.2.2.4	What metadata is needed?	10
2.2.2.5	What interoperability and integration problems exist?	19
2.2.3.1	How can data be integrated (e.g., channel integration, functional integration, data warehousing, database federation)?	22
2.3.2.1	What are the economic costs associated with DIQ in an information system, database, or accounting system?	10
3.1.1.1	Who decides what data quality dimensions and metrics are relevant?	11
3.1.2.1	What data quality dimensions and their corresponding metrics exist or should be monitored?	80
3.1.2.2	What is the relationship between how data is used and data quality dimensions and metrics?	24
3.1.2.3	What procedures, surveys, and metrics must be in place to validate and verify DIQ?	12
3.1.2.4	What is DIQ?	29
3.1.3.1	How is fitness-for-use defined?	10
3.1.3.2	How does quality differ by stakeholder?	38
3.1.3.3	How are data quality dimensions measured and reported?	16
3.1.3.4	How should data quality metrics be used to detect and correct errors in data?	10
3.1.3.5	How good is good enough, with respect to data quality?	11
3.1.3.6	How does data quality vary as a function of time?	10
3.2.2.1	What should be included in systems design to ensure DIQ?	34
3.2.2.2	What new technologies and processes are available to manage data uncertainty (e.g., approximate, probabilistic, inexact, incomplete, imprecise, fuzzy, inaccurate data)?	11
3.2.2.3	What are the features of DIQ tools?	21
3.2.3.1	How are systems' design processes modified to include DIQ elements?	15
3.2.3.4	How does the system architecture affect DIQ?	10
3.2.4.2	What costs (e.g., poor decision-making, increased legal costs, and bad public relations) are associated with not incorporating DIQ into systems design?	16
3.2.4.4	What benefits are associated with incorporating DIQ into systems design?	19
3.3.2.1	What conceptual models or matrices exist (e.g., IP Maps, Kano)?	25
3.3.2.2	What semantics notation is needed to define IP?	11
3.3.3.1	How are conceptual models used to improve DIQ?	13
3.3.3.2	How are semantic notations used to represent DIQ issues?	21
	Total	649

health care providers prove useful in improving the quality of care delivery in some settings and in motivating documentation in others." See Table 7 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Governance Strand—Legal and Ethical Compliance (Section 1.2)

Compliance with laws, regulations, and ethical codes of conduct is increasingly important. With changes on the horizon related to industry-specific regulations, as well as conformance with GAAP and IFRS, the accounting field will increasingly play an important role in operations. Legal and ethical considerations are clearly an issue in the accounting domain. Sarbanes-Oxley (SOX) requires management to make assertions about internal control. Stockholders and other stakeholders can be harmed financially if corporations do not comply with GAAP, IFRS, or SOX. Researchers and practitioners are concerned with both the economic and noneconomic impact of compliance with these existing laws and regulations. With the questions in this research strand, safety issues are also examined. Safety issues, with respect to the accounting domain, principally concern privacy and financial security (of investors). Within the broader DIQ literature, there is the question: "How does non-compliance harm people (1.2.5.3)?" Fisher and Kingma (2001) discuss the role of DIQ in the Challenger and USS Vincennes disasters. Death is the most serious result of inaccurate, incomplete data coupled with information overload, experience level, and time constraints. None of the questions in this section appeared in the top 20 percent of researched questions. See Table 8 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Governance Strand—Personnel Management (Section 1.3)

It is important to have personnel in place that are aware of DIQ issues and are trained to recognize inadequately implemented DIQ policies and procedures. Questions in this research strand are concerned with making employees aware of the importance of DIQ through training; motivating them to comply with DIQ policies and procedures and measuring employees' DIQ achievements; and with understanding the economic and non-economic consequences of noncompliance.

The personnel, who develop and implement AIS, as well as the end-users of the systems, should be trained and provided incentives to ensure high quality data. Hence, an important question in this research strand lies in "What motivates employees to care about DIQ (1.3.2.3)?" Within the financial services area, Klein (1997) examines how actuaries detect and correct data errors, stating, "The extent to which attempts to find errors are pursued appears to be driven by a trade-off between the time and effort required to find additional errors and the potential impact of a more accurate dataset." None of the questions in this section appeared in the top 20 percent of researched questions. See Table 9 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Operations Strand—Production (Section 2.1)

Research questions relating to production generally deal with the way the data flows through the system, eventually becoming an IP. Accounting information systems are ultimately production processes, collecting transactional data and converting it into IPs such as financial statements and managerial reports. Production processes deal with the quality requirements in the procurement of raw data, data structures, and records needed for the production of IPs. The production system is the process that ensures that the transformation of data into information is correct (as opposed to the correctness of the data itself) utilizing, for example, data tags and extract-transform-load (ETL) tools. It is concerned with the identification of non-conforming data items and specifications of corrective action.

With 40 papers addressing this subcategory, many of the topics for this area have been extensively researched. This is not to say opportunities for research in this area do not exist, but the authors suggest that anyone examining these questions should ensure their approach is unique or extends existing work in a meaningful way.

Six questions from this subcategory were in the top 20 percent of questions most frequently researched (see Table 2). The focus of these questions is data integrity, data cleaning, and system interventions to improve the quality of data. Thus, the system should "do no harm" as the data is processed through it. Becker (1998) noted that the need for data quality improvement initiatives will continue to grow as databases become more complex due to technological advances. This has proved to be the case with the explosion of data and information found in corporate intranets and extranets, as well as the Internet in general.

The question of "What is data integrity (2.1.2.3)?" has been well researched. Lee et al. (2004) define it with constraints and objectives in their paper on embedding data integrity into a continuous data quality improvement process. They propose that the application of data integrity be viewed as a "dynamic, continuous process, embedded in an overall data quality improvement process (87–88)." In addition, several articles deal with "How is data cleaning achieved (2.1.3.2)?" Thiru et al. (1999), Orman (2001), Zhu and Wu (2004), and Winkler (2004) all describe various methods of cleaning data. See Table 10 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Operations Strand—Distribution (Section 2.2)

As with any manufacturing process, the product (in this case IPs) must be distributed. The questions in this section deal with moving data through the system as well as data integration, metadata, and quality documentation and records. When physical products move through the distribution system, they seldom change. However, as data moves away from the original source, the meaning of the data can change.

Managerial accountants should help determine what metadata is appropriate to capture in order to ensure that IP adequately supports decision-making. As data moves from traditional accounting systems into secondary sources such as decision support systems and data warehouses, the collection of this metadata becomes more critical. Contextual data and other data tags may not be necessary with respect to its primary purpose but may be quite important in secondary systems.

Documentation is an important part of any system development project. The development of AIS includes entity relationship diagrams (ERDs) or Resources Events Agents (REA) models. Within the DIQ literature, several alternative modeling tools are proposed: IP Maps (Shankaranarayanan et al. 2000) and IASDO model (Thi and Helfert 2007); modified IDEF0 (Al-Hakim 2008); and IQ Matrix, IQ Unit Identification, Task IQ tree, IQ network (Wang and Wang 2008). Modeling is a promising area for future research in accounting. In addition, the incorporation of metadata at the transaction processing level can help to support data as it moves into secondary sources that are used to support decision-making.

Distribution is a widely researched area. The authors reviewed 27 papers with a primary focus of distribution. The papers are in government, financial services, and multiple theoretical areas. Not only is there a variety of papers in regards to this section, but many different questions in this section are addressed. Three of the top 20 percent of questions are found in this research strand. These questions are primarily focused on meta-data and integration. Although much of the existing research is technical in nature, the accounting field can contribute by identifying meta-data that is appropriate to decision-making using data from secondary sources, as well as looking at resolving issues of integration based on predefined taxonomies such as XBRL. See Table 11 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Operations Strand—Operations & Assurance Services (Section 2.3)

Questions in this research strand are highly relevant to accounting and AIS. It is also an area in which the accounting discipline can substantially contribute to the DIQ literature. Operations and assurance services pertain to the services provided by an independent party to validate or improve DIQ or its context, including prevention, appraisal, and failure costs. There are questions related to the demonstration and proof of quality as required by customers and management, including such services as business risk assessment, information systems security review, customer satisfaction surveys, internal and continuous auditing, and accounts receivable review.

As discussed earlier, the authors reviewed 294 articles in JIS and IJAIS. Within these articles, 35 had the word "assurance" in their title or abstract and 74 articles had the word audit. Frequently, these words are considered synonymous with quality. Yet, only a small number of papers dealt specifically with DIQ. The authors discovered only nine articles that deal with assurance services. This suggests that there is a great deal of opportunity to do further research in DIQ that is directly related to AIS.

One research question in the top 20 percent appears from this group: "What are the economic costs associated with DIQ in an information system, database, or accounting system (2.3.2.1)?" Even and Shankaranarayanan (2007) propose a framework for modeling and quantifying some economic effects of quality configurations. Others simply acknowledge that there are economic costs associated with DIQ but do not propose solutions.

Another research question that has been addressed is: "Who is responsible for assurance services (2.3.4.1)?" As might be expected, the answer is auditors (Kaplan et al. 1998; Debreceny et al. 2005; Lala et al. 2002). Assurance services are a core function of both internal and external auditors. Stakeholders increasingly want to have accurate, timely, and relevant financial information to support decisions such as investing in stock and expanding a business. New processes such as continuous auditing and software to support the audit function are allowing the data and information released to stakeholders to have some assurance associated with them. Along with this benefit, there are costs related to new systems and personnel. Further research in this area is needed to ensure that the costs and benefits can be adequately measured in monetary and human elements. See Table 12 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Technology Strand—Input: Dimensions & Measurement (Section 3.1)

The technology sections of PGOT generally deal with the standard system areas of AIS: input, systems and tool design, and output. Within DIQ, input issues are related to dimensions, measurement/metrics, and fitness-for-use. Research questions are concerned with getting the data right at the source and determining if the data presented to the consumer will help in the decision-making process.

Data is expected to be accurate in many domains, but particularly in the accounting domain. The focus of much of the DIQ literature is on the determination of what actually constitutes DIQ, and the conclusion is that DIQ is multi-dimensional. The accounting and AIS fields can benefit from this research and begin to implement more comprehensive measures to ensure that data and information are of the highest quality for the task at hand.

The dimensions and measurement subcategory has been highly researched; particularly in the MIS domain. In particular, probably the most studied topic in the DIQ literature, the question is: "What data quality dimensions and their corresponding metrics exist or should be monitored (3.1.2.1)?" Wang and Strong (1996) provide the comprehensive list used in the introduction to this paper. There is agreement from other researchers concerning this list (Al-Hakim 2008; Klein 2002; Divorski and Scheirer 2001). This list, however, does not address the fact that these varying attributes of data quality involve trade-offs when used in a decision-making context. As indicated earlier, much of the DIQ literature is focused on what constitutes high quality data. The most important contribution of this earlier work to the AIS literature is the recognition that quality is more than accuracy. Automated tools can help to ensure the accuracy of accounting data, but it takes human intervention to determine the relevancy of the data to a particular decision. As AIS components become deployed on intranets and the Internet, security and accessibility become ever more important. Unfortunately, the accounting domain currently has to deal with issues of believability, given recent corporate scandals and disasters. All of these are considered dimensions of DIQ.

The authors reviewed 35 articles that had a primary focus of technology and a subcategory of input. Most of these papers are theoretical in nature and are found in the traditional DIQ journals. Eleven of the top 20 percent of questions most frequently researched were from this subcategory (see Table 2). The "What" questions have been well addressed (e.g., what are the dimensions of quality, what trade-offs exist, what dimensions should be monitored). More difficult questions pertaining to "How" to accomplish these goals is still an area for further research. See Table 13 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Technology Strand—Systems and Tool Design (Section 3.2)

This subcategory deals with incorporating data quality into the design of an information system, data quality tools, and metadata tools. Here, the goal is to create a quality system that will produce quality data. As AIS become increasingly more comprehensive (e.g., systems such as SAP), it is more important to include data quality and metadata tools as part of the system design. In addition, the cost of not paying attention to what is happening in the "black box" can be substantial. The DIQ literature is relevant to the accounting domain in highlighting ways to ensure that the "black box" is more transparent. It is the opinion of the authors that the current focus in the DIQ literature on data cleansing and metadata tools can provide context for incorporating these tools into AIS. In particular, the recognition and implementation of data quality can allow external auditors to rely more on the system, thus reducing the amount of substantive testing that needs to be done.

The authors reviewed 17 papers with a primary focus of systems and tool design, most of which are primarily theoretical in nature. Although accountants should be concerned about the "black box," this is an area of less concern to the accounting literature and should remain a concern of technical fields such as computer science, software engineering, and MIS. See Table 14 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Technology Strand—Output: Information Products (Section 3.3)

IP, which is essentially system output, takes on many forms such as a report, invoice, or chart. The AIS literature can contribute to DIQ by focusing on the quality of the many ways IPs are generated as it directly relates to the decision-making context. Organizations interested in DIQ should designate an IP manager. This is someone who coordinates and manages suppliers of raw information, producers of deliverable information, and interfaces with information consumers (Pierce 2004). Thus, issues in this research strand pertain to the analysis and design of the quality aspects of IPs as well as the semantics of the output.

As indicated in the literature, there are several models useful in the analysis and design of IPs including IP Maps (Shankaranarayanan and Cai 2006); Control Matrices as used by IS auditors to evaluate how reliably a system safeguards assets and protects data integrity (Pierce 2004); and the Kano model, which can be used as a framework or method for identifying quality expectations and the time transition of quality factors (Zhang and von Dran 2001).

Models have been well developed in the MIS field and, given the nature of IP, should not need adjustment to use in the accounting domain. Four questions in this research strand, primarily dealing with models and semantics, are in the list of top twenty percent of questions. This is probably not an area that needs to be heavily researched in the accounting domain.

Much of the literature on IP focuses on reports developed by IT or MIS staff. As MIS departments become overwhelmed with information requests, more of the reporting is controlled by the end-user. This may lead to end-users extracting correct data in incorrect ways. For example, the uninformed end-user may create a Cartesian product, which matches unrelated records, resulting in information that is incorrect even though the underlying data is accurate. Careful design of systems can help to alleviate this problem. Debreceny and Bowen (2005) investigate the relevance of including object-oriented (OO) features such as generalization-specialization hierarchies (GSHs) and abstract data types (ADTs) within the design of relational databases, in order to facilitate end-user queries. Their findings show that these OO features result in fewer semantic errors and shorter query formulation time. Thus, incorporating OO features into relational databases may lead to more reliable information.

In two articles, (Dunn and Grabski 2000, 2001) explore the cognitive issues associated with IPs. Their research experimentally shows that users perceive REA (Resources Events Agents) as more semantically expressive than the traditional Debit-Credit-Account (DCA) accounting model. Controlling for cognitive fit, accounting knowledge, and field dependence, higher perceived semantic expressiveness is associated with higher task accuracy. Thus, better decisions can be made. In addition, the localization of the relevant objects or linkages is important in establishing cognitive fit (i.e., resources, events, and agents versus debits, credit, and accounts). See Table 15 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

Decision-Making Strand (Section 4)

In the long run, an AIS, or an information system in general, must support decision-making. This research strand focuses on decision-making in context, or fitness-for-use. This research strand does not have any subcategories. It is the culmination of all of the previous strands and takes input from each one. It should be noted that this research strand does not encompass all aspects of decision-making. It focuses on the impact that the quality of data and information has on a decision outcome.

Ultimately, one of the most difficult questions in DIQ is: “How do end-users determine if DIQ is sufficient for the decision being made?” This query leads to other questions. How do managers use the data and information that are available to them to plan operationally, tactically and strategically? What cognitive skills enable end-users to make better decisions? What dimensions of quality are important in the decision-making process? Is it always true that accuracy is more important than timeliness? Or are there circumstances where “mostly” accurate data will suffice if it is received in a timely manner?

Many of the accounting oriented papers that were reviewed fell in this research strand. This should not be surprising since a primary purpose of AIS is to provide managers with the products to support decision-making. As organizations become more complex, it is increasingly important for systems to provide data beyond the traditional debits and credits. Managers need information about trends to create sales and production schedules. They must be able to accurately track costs so that pricing can be effective.

An important aspect of this strand is the ability of the end-user to extract the data from the system in a way that will answer the question asked. Data may be accurate in the system, but if the end-user is unable to extract it in a meaningful way, then the quality of the information is insufficient to address the decision maker’s needs. Text-based interfaces are more effective for low complexity tasks and a visual interface is more helpful when task complexity is high (Speier and Morris 2003). When end-users translate natural language queries into pseudo-SQL they remove some of the ambiguities in the query, thus increasing the probability that queries will return the correct information for the task at hand (Borthick et al. 2001; Bowen et al. 2004). Further, the structure of the database (i.e., first normal form versus third normal form or relational database versus object-oriented database) may impact the ability of the end-user to effectively extract the necessary information (Borthick et al. 2001; Bowen and Rohde 2002; Bowen et al. 2006). In order to adequately address decision-making in context it is important to consider these issues.

The unspoken goal of all previous DIQ work is to enhance decision-making in context. Decisions are made at all levels of an organization and the IP that support these decisions can be as simple as a quarterly sales report or as complex as an analysis of sales trends over several years, several products and with comparisons to sales of other companies in the same industry. With this in mind, questions in this strand should strive to address the complex questions posed earlier. This continues to be an area where accounting and AIS researchers can make a significant contribution. See Table 16 of the online resources (http://aaapubs.org/) for a complete list of articles pertaining to this topic.

IV. FUTURE RESEARCH DIRECTIONS

The majority of the 193 papers reviewed are in the MIS domain (91). Other domains represented within the population are management (32), accounting (23), software engineering (21), computer science (13), operations research (9), and library science (2). Although only 23 articles have been identified as within the accounting and AIS domain, looking at DIQ papers in general can provide direction for addressing the same or similar questions from an AIS or accounting perspective. The previous two sections contain numerous observations about what has and has not been addressed with respect to specific research questions. When appropriate, interpretations of the findings in other domains were made above with respect to accounting in order to identify specific research questions that may be worthwhile to pursue. The previous sections detailed the DIQ literature that has been reviewed spanning from 1994 to 2008. The current section will provide more general observations. Specifically in this section the authors discuss research strands that have been under-studied and offer opportunities for further research, particularly as it relates to accounting, auditing, and accounting information systems.

Each of the 193 articles were coded with a primary PGOT and fitness-for-use focus (e.g., personnel management, how). This provides a broad picture of the existing DIQ research field. For a more granular analysis each of the 193 articles was examined with respect to the 173 developed research questions. Since these questions were generated with respect to the PGOT framework, the questions facilitate classification of the existing research within the framework. This coding provides evidence that a particular article may examine multiple questions, albeit in minor ways. Thus, we can see both the breadth of the existing research and the depth. This comparison is shown in Table 3.

Although it is interesting to reflect on the overall statistics, it is more fruitful to look at areas that have been under-researched. Thirty-three potential research questions exist that have not been addressed in any of the reviewed articles (see Table 4). Although these questions are generated by the researchers, they are reflective of their expertise and indicative of questions that should be addressed in the field.

In comparing Table 4 with Figure 3, it is evident that the 33 questions that have not been addressed in any of the literature are concentrated in a few areas. Governance questions comprise 36 percent of the questions (15 percent overall are Legal and Ethical and 21 percent overall are Personnel Management). Of the 39 percent of the questions related to Operations, 21 percent are classified as Operations and Assurance Services. Of the remaining questions, 6 percent are Production and 12 percent are Distribution questions. Finally, with respect to Technology, 3 percent are related to Input, 15 percent are Systems and Tool Design questions, and 6 percent are Output, for a total of 24 percent. Clearly, the biggest gaps are in Legal and Ethical, Personnel Management, and Operations and Assurance Services. These are critical areas in the accounting, auditing, and AIS domains and will be discussed individually.

TABLE 3

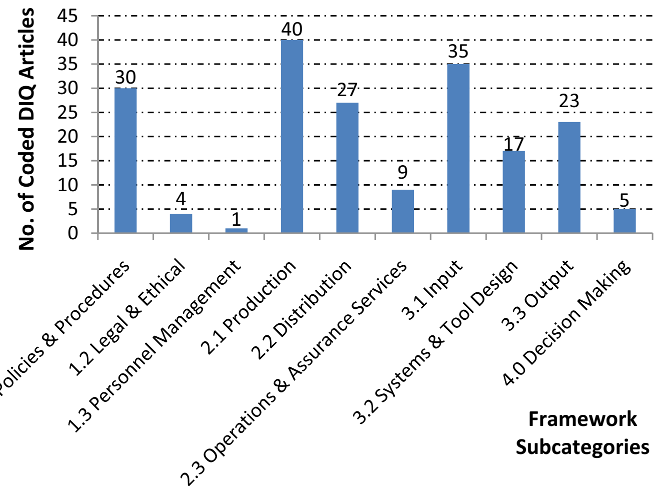
Primary PGOT Research Focus Compared to Total Number of Times a Research Question Was Addressed for the 193 Coded DIQ Articles

Research Strands		Fitness-for-Use Categories: Who		What		How		Econ. Resource		Safety		Totals	
Categories	**Subcategories**	**x.x.1.x**		**x.x.2.x**		**x.x.3.x**		**x.x.4.x**		**x.x.5.x**		**Prim.**	**All**
Governance	1.1 Plans, Policies and Proc.	0	12	25	38	3	15	2	18	0	9	30	92
	l.2 Legal and Ethical	0	7	2	7	2	2	0	3	0	4	4	23
	1.3 Personnel Management	0	11	0	13	0	9	0	0	1	3	1	36
Operations	2.1 Production	0	11	18	42	21	78	0	5	1	10	40	146
	2.2 Distribution	0	5	12	60	15	37	0	8	0	8	27	118
	2.3 Operations and Assurance	0	1	3	21	2	15	4	15	0	0	9	52
Technology	3.1 Input	0	16	20	145	15	95	0	4	0	5	35	265
	3.2 Systems and Tool Design	0	5	7	66	10	39	0	40	0	3	17	153
	3.3 Output	0	9	11	52	12	41	0	8	0	4	23	114
Decision-making within context		0	0	2	5	5	7	0	0	0	0	7	12
Totals		0	77	100	449	85	338	6	101	2	46	193	1011

TABLE 4
Research Questions Not Addressed by Any of the 193 Coded DIQ Articles

Number	Question
1.2.1.3	Who is responsible for ensuring DIQ practices conform to the corporation's ethical code of conduct?
1.2.3.2	How should internal controls be developed?
1.2.3.3	How are ethical dilemmas handled?
1.2.4.1	Who estimates the cost of legal and regulatory compliance?
1.2.5.1	Who determines whether poor DIQ will harm individuals, groups, organizations, or society?
1.3.1.2	Who should train collectors, custodians, and consumers concerning DIQ policies and procedures?
1.3.3.2	How do you assess the effectiveness of DIQ training and incentives?
1.3.4.1	Who decides how money is spent to motivate and train employees?
1.3.4.2	What is the economic impact when personnel are improperly trained, resulting in poor DIQ?
1.3.4.3	How much money should be allocated to DIQ training and the incentives budget?
1.3.5.1	Who determines whether poor DIQ training will cause harm to people?
1.3.5.3	How should training be conducted to ensure DIQ does not harm people?
2.1.1.2	Who is responsible for assigning data tags (e.g., information about the quality of the data)?
2.1.4.2	What are the cost categories and benefits of production, including data cleaning and integrity checks?
2.2.1.3	Who is responsible for data integration?
2.2.2.7	What processes exist to prevent misdirecting or distributing information to unauthorized people?
2.2.3.5	How often should data be integrated and refreshed?
2.2.4.1	Who decides whether an investment in metadata is made and if so, what tools are purchased?
2.3.1.2	Who defines the metrics for measuring operation and assurance costs?
2.3.1.3	Who determines the budget for data quality enhancement?
2.3.2.5	What role do internal auditors play in determining and evaluating security controls for protecting sensitive data?
2.3.3.1	How are DIQ costs gathered and measured?
2.3.5.1	Who within the company implements recommendations provided by third parties?
2.3.5.2	What are the operation and assurance costs associated with human safety?
2.3.5.3	How do assurance services minimize the negative impact on people?
3.1.4.1	Who determines what the budget is for measuring data quality?
3.2.1.2	Who identifies emerging technologies and best practices for managing DIQ?
3.2.1.3	Who identifies potential DIQ tool vendors and is responsible for evaluating such tools?
3.2.3.2	How does compliance with GAAP and IFRS impact DIQ systems design?
3.2.4.1	Who approves the funding of DIQ systems development, including the purchase of DIQ tools?
3.2.5.3	How does one prevent unauthorized access to the system (e.g., to prevent identity theft and financial fraud)?
3.3.4.1	Who determines what money is available to fund improvements in the quality of IP?
3.3.5.2	What measures are used to show the impact of DIQ with respect to how IP affects human safety (e.g., percentage of loss on retirement accounts or investment decisions)?

FIGURE 3
Primary PGOT Subcategory Focus for the 193 Coded DIQ Articles



Breaking these 33 questions down by fitness-for-use categories, three-fourths of the questions are related to "Who" (24 percent), "Economic Resources" (27 percent), and "Human Safety Issues" (24 percent). The "What" and "How" questions have been well addressed in the existing literature. Opportunities exist, particularly with respect to "Who" questions, for the AIS and accounting domains to contribute significantly to the DIQ literature.

Legal and Ethical Compliance

Scandals in business, including the disintegration of the sub-prime mortgage market and business collapses such as Enron, have brought legal and ethical issues to the forefront of the public eye. However, much of the literature related to accounting ethics deals with the ethics of accounting students (Blanthorne et al. 2007; Guffey and McCartney 2008). As a result of the accounting scandals of the late 20th century, Sarbanes-Oxley Act of 2002 provides legal ramifications for misrepresentation of financial statements and requires management to provide an assessment of internal controls. The quality of data and information is an implicit requirement of the financial statement presentation. Thus, an area of opportunity for research is at this junction of the legal requirements and the quality of the information presented to stakeholders. The consideration of ethical ramifications of the way data is presented (a dimension of DIQ) is another area that would be both useful and informative.

Two questions will increasingly be important from a legal and ethical perspective: (1) What legal and regulatory requirements exist that pertain to DIQ (e.g., GPRA, GAAP, and IFRS) (1.2.2.1)? and (2) How is compliance with legal requirements accomplished (1.2.3.1)? Regardless of where organizations do business, they will be expected to comply with local laws and regulations. The fitness-for-use constructs developed in this paper highlight the role that DIQ plays in this arena. International organizations may deal with conflicting regulations, thus necessitating trade-offs among the various DIQ dimensions. Accountants can facilitate the process of reconciling conflicting regulations and embed these in AIS, thus automating the process.

Personnel Management

The DIQ literature as reviewed in this study is primarily concerned with training people on finding errors, as well as creating new positions within organizations to create, manage, and monitor DIQ policies. A review of job descriptions on various job-posting sites reveals that accounting knowledge is not necessarily a prerequisite for these types of positions. An opportunity exists to review the qualifications for these positions and propose new methods of identifying the best candidates to fill the positions.

The role of DIQ manager requires an understanding of the entire organization and individuals with accounting knowledge are well suited to fill these jobs. They are accustomed to analyzing data and looking for patterns and discrepancies. In addition, they have knowledge of business processes, giving them perspective across the organization. As businesses move to integrated platforms such as ERP systems and data warehouses, the ability to gauge quality outside of traditional silos becomes increasingly important. With this broad organizational outlook, accountants can help to define policies and procedures, as well as metrics for measuring compliance.

Another question in this area is concerned with who should train collectors, custodians, and consumers concerning DIQ policies and procedures? As accountants and auditors routinely work with individuals along the entire lifecycle of information, they are in a position to develop training programs to make people aware of the costs of poor DIQ. Ideally, organizations will have a data quality team in place, consisting of individuals who would be responsible for training and tracking compliance. Thus, although the accountants and auditors would be fundamental in developing policies, plans and metrics, they would be implemented by individuals who are qualified trainers. These individuals would have authority to reward compliance with policies and procedures and punish for non-compliance.

Operations and Assurance Services

There is opportunity for accounting and AIS researchers to contribute to the DIQ literature in this area. The AICPA and CICA have developed the Trust Services Principals and Criteria guidelines for providing assurance on IT-related systems resulting in WebTrust® and SysTrust® (Boritz and Hunton 2002). These services are provided by public accounting firms and indicate a willingness of the accounting profession to pursue additional assurance services. Given that auditing is within the domain of accounting, DIQ audits are a natural extension. Within the area of Assurance Services, it is important that the literature move beyond accuracy and other intrinsic data qualities and strive to examine the DIQ on multiple dimensions. The ability to audit the completeness, accuracy, validity, and accessibility of data and the information technology that supports the data is well represented within CobiT. However, auditors and accountants are in a position to determine other dimensions of DIQ including timeliness, believability, representational consistency, and relevancy. These representational and contextual dimensions are vital to arriving at DIQ that is fit for use. New audit programs that incorporate these dimensions when evaluating financial data should be developed.

Fitness-for-Use

Reviewing Table 3 from above, it is obvious that, with respect to fitness-for-use, the areas of "Who," "Economic Resources," and "Human Safety" have been under-studied. The obvious answer as to "Who" is responsible for DIQ, particularly within the accounting domain, would be auditors, either internal or external. However, research opportunities exist to identify other personnel who could assist in this process. What skills and education would be necessary to ensure that data is of the highest quality? Should accounting curriculums explicitly include DIQ as an element of coursework? When new auditors are hired, should the impact of DIQ on an organization be a part of the training? Additionally, the role could be technical, involving changes to the AIS, or the role could be analytical, involving IPs and the implementation of procedures to determine the DIQ.

Much of the literature related to economic resources pertains to model development for allocating the costs associated with data quality enhancement. Implementation of some of these models in accounting information systems could support continuous auditing and the facilitation of cost allocation models. With respect to human safety issues, the biggest concern would be in the areas of data privacy as well as assurance seals such as WebTrust®.

Some Research Opportunities

Given the above discussion regarding large areas of DIQ that have been under-studied and research methods that have been underutilized, the authors now propose several possible research areas within the larger PGOT framework worthy of study. As summarized in Table 5, there are several opportunities for theory and model building. Especially relevant, given the propensity for experimental research in the AIS field and the paucity of it in the traditional DIQ literature, is the possibility of testing many of the theories developed in the earlier DIQ literature.

People

Within the people element, further research is needed to develop and test theories that will provide guidance for responsibilities within the IP lifecycle. Within the accounting domain, collectors, consumers, and auditors all play a key role. More significantly, collectors can be directly responsible for the accuracy of data, while consumers are directly impacted by the context in which the data and information is presented. Auditors play a considerable role in the assurance of DIQ. They are concerned with many aspects of DIQ; accuracy, consistency, security and reliability to name a few. As auditors move more into management responsibilities, it may be time for them to consider other aspects of DIQ such as accessibility or ease of understanding.

Organizations employ accounting personnel in a variety of roles: financial, managerial, and auditing. In addition, accountants should be involved in the development of AIS and decision support systems for organizations. With the ability to collect input from many areas of the business, accountants are in a unique position to develop and test hypotheses that evaluate the importance of different people (e.g., collectors, custodians, consumers, and auditors) in the information life cycle. These hypotheses should focus on assuring that DIQ is sufficient to support decision-making. The following questions could be answered via a case study or survey method:

- Who should be responsible for approving DIQ-related plans, policies, and procedures?
- Who should approve funding of systems and improvements to ensure DIQ?
- Who controls the budgetary resources and manages constraints associated with DIQ initiatives?
- Who is responsible for compliance with laws and regulations and auditing internal controls that concern DIQ?

Another area that needs exploration is the development of theories to explain how people evaluate the importance of DIQ within the information lifecycle (e.g., collectors may not see DIQ as important whereas consumers may see it as critical). These theories can then be used to alter behavior so that DIQ can occur at all stages of the lifecycle. Experimental protocols, particularly within industry (as opposed to student populations) could prove or disprove these theories. A related question is predicting the impact that training will have on people at different lifecycle points. Action research is well suited to this question. Working within an organization, the predictions can be tested by altering the environment, training individuals, and tracking the outcome of the changes.

TABLE 5

Research Opportunities

PGOT Element	Some Research Opportunities
People	• Develop and test hypotheses to evaluate the importance of different people (e.g., collector, custodian, consumer, auditor) in the information life cycle in assuring that DIQ is sufficient to support decision-making ◦ Who should be responsible for approving DIQ-related plans, policies, and procedures? ◦ Who should approve funding of systems and improvements to ensure DIQ? ◦ Who controls the budgetary resources and manages constraints associated with DIQ initiatives? ◦ Who is responsible for compliance with laws and regulations and auditing internal controls that concern DIQ? • Develop theories to explain how people evaluate the importance of DIQ within the lifecycle (e.g., collectors may not see DIQ as important whereas consumers may see it as critical) • Predict the impact that training will have on people at different life cycle points
Governance	• Using longitudinal case studies, evaluate the effect of plans, policies and procedures on the DIQ of an organization • Predict how changes in DIQ plans, policies and procedures will improve DIQ and test it empirically • Evaluate how legal and ethical considerations are affecting the need for better DIQ and the resources available to meet regulations • Create cost/benefit models that can be tested empirically for all aspects of governance including training and incentives for creating quality data in the first place • Evaluate the effect of training on the desired outcome of improved DIQ • Evaluate the effect of poor DIQ on human health and safety • Using theory from reference disciplines, develop frameworks for handling ethical DIQ dilemmas • Evaluate the impact of DIQ in complying with IFRS
Operations	• Evaluate the costs of incorporating data quality information (data tags) into metadata • Evaluate the costs of data quality attribute metrics and incorporating them as data tags • Develop and test hypotheses that explicitly incorporate DIQ in the development of source data that will be used in secondary sources • Determine how data quality tools can be used most effectively to improve DIQ • Develop models of data integration to guide policy making decisions • Develop metrics for evaluating the cost of poor DIQ on human safety
Technology	• Evaluate alternative system development models (e.g., REA) with respect to explicitly including DIQ as a system requirement • Develop knowledge-based tools that will supplement human intuition • Evaluate existing technologies and best practices for managing DIQ • Develop cost/benefit models for technology that explicitly addresses DIQ issues • Predict the characteristics of IP that will allow for the most versatility (e.g., can be used by the most decision makers in different contexts)
Decision-Making	• Determine cognitive characteristics and abilities that are most relevant in assessing DIQ in context • Empirically evaluate how decision makers select which among the many competing data quality dimensions are most important • Predict which dimensions of quality are most important in specific decision-making contexts (e.g. accuracy, security, or representational consistency)

Governance

The accounting domain is equally suited to exploring governance-related issues. For example Wilkin and Chenhall (2010) review IT governance and their taxonomy addresses the technical aspects of accounting information systems. Within the scope of the PGOT framework, governance is related to polices, plans and procedures, legal and ethical compliance, and personnel management. Although legal and ethical issues have been studied in the field, it would be interesting to consider governance concerns with explicit DIQ considerations. In addition, the accounting field is well positioned to evaluate the impact of costs and benefits of implementing DIQ plans, policies, and procedures on the IPs that are generated. Accountants tend to be detail-oriented and auditors use judgment in decision-making. How can these abilities impact the DIQ of an organization? Studies that explore this question can ultimately lead to the development of more effective systems with better controls.

There are several areas that need further evaluation including:

- Evaluate the impact of DIQ in complying with IFRS
- Evaluate how legal and ethical compliance affect the need for better DIQ and the resources available to meet regulations
- Evaluate the effect of training on the desired outcome of improved DIQ
- Evaluate the effect of poor DIQ on human health and safety

Each of these areas ultimately impact the DIQ of financial statements and other reports used in management decision-making, or the ability of this data and information to be used effectively. Studying these issues will provide foundations for future work.

Using longitudinal case studies, researchers could evaluate the effect of plans, policies, and procedures on the DIQ of an organization, with an emphasis and focus on how DIQ changed as a result of the plans, policies, and procedures. Using the longitudinal approach allows testing of DIQ at multiple stages. It would be important to have consistent measures at the various testing points. Development of metrics would facilitate the process. Alternatively, the impact of DIQ on the decision-making process could be measured, via a survey or interview process. A related opportunity is to predict how changes in DIQ plans, policies, and procedures will improve DIQ and test it empirically using an experimental approach.

Particularly in light of recent corporate scandals the accounting discipline could benefit from using theory from reference disciplines to develop frameworks for handling ethical DIQ dilemmas. This can be done in conjunction with the overall examination of ethical dilemmas that the profession is dealing with. Another area in which accounting and AIS researchers can contribute would be the creation of cost/benefit models that can be tested empirically for all aspects of governance including training and incentives for creating quality data in the first place. With their attention to detail and knowledge of relevant costs this is an area in which the discipline could make a significant impact.

Operations

Operations have traditionally been studied more from an MIS perspective. However, the input of the accounting field can yield many benefits. In particular, DIQ metric definition and measurement are lacking. Due to the fact that accountants are generally oriented toward metrics and measurements, they may be able to use this prior knowledge and experience to develop new ways of creating DIQ metrics in their field. In addition to metrics, data quality tags and metadata have the potential to improve DIQ.

Providing information to end-users about the quality of the data in the reports they receive may be perceived as information overload. Although this has been studied in MIS (Fisher et al. 2003), the study involved decision-making with respect to choosing an apartment. Extending this experiment by including metadata with financial statements would generalize the findings. In addition, this work could be followed empirically by studying the impact on earnings and stock prices. There is a fine line between providing decision makers with the information needed to determine the quality of the information they are relying on and information overload. Providing guidance in this area would be a significant contribution.

In order to improve decision-making, data quality information (data tags) can be added as metadata at the system level. As an example, XBRL General Ledger is a taxonomy that allows disparate financial systems to talk to each other in a common language (http://www.xbrl.org/GLTaxonomy). This metadata is added to the underlying financial data using software tools such as those developed, for example, by Clarity Systems or EDGAR Online, Inc. Other examples of providing data tags would include adding explicit quality attributes at the record or field level. However, there are costs associated with incorporating data quality information into the metadata. These costs should be evaluated in terms of information overload and dollar cost. A careful evaluation of these costs will be an important step in controlling the costs. Additionally, it would be interesting to examine the data quality dimensions that deal with intrinsic DIQ (i.e., accuracy, objectivity, believability, and reputation) and study the costs of developing metrics and incorporating them as data tags. A related area of study lies in the development and testing of hypotheses that explicitly incorporate DIQ in the development of source data that will be used in secondary sources. Again, as accountants are routinely involved in the decision-making process that uses these metrics, it is an area in which the discipline can contribute. It is also an area where the discipline would benefit by bringing the issue from an implicit concern (DIQ is a problem) to an explicit concern (incorporating DIQ information could result in better decision-making).

Some areas of further research that allow for more interdisciplinary work are as follows:

- Determining how data quality tools can be used most effectively to improve DIQ
- Developing models of data integration to guide policy-making decisions
- Developing metrics for evaluating the cost of poor DIQ on human safety

Although much research in operations has been done in the MIS domain, it is clear that opportunities are available within the accounting and auditing domain. This is an area rich with opportunity for interdisciplinary work, building on what has already been proposed.

Technology

Like operations, this is an area traditionally studied in MIS, computer science, and software engineering. However, there are areas that are relevant to the accounting discipline and where accounting and AIS would be able to contribute. As indicated earlier, the accounting discipline has already developed an alternative system development model (e.g., REA). Extending the model to explicitly include DIQ within resources or events may improve the ultimate efficiency of AIS.

An interdisciplinary opportunity exists for the development of knowledge-based tools that will supplement human intuition. Knowledge-based tools can use fuzzy logic and complex algorithms, traditionally the domain of computer science. However, developing metrics and heuristics that accountants and business people use in decision-making could augment these models.

Finally, predicting the characteristics of IP that will allow for the most versatility (e.g., can be used by the most decision makers in different contexts) is an area where the domain knowledge of accountants can be leveraged. As indicated previously, accountants occupy many of the positions throughout the lifecycle (consumers, collectors, auditors) and can provide insight into the appropriateness of various reports and other information products.

Decision-Making

As previously noted, much of the reviewed accounting literature in the DIQ field is focused on decision-making. However, there is still much opportunity for additional work in this area. One area that is a natural extension of previous work is to determine the cognitive characteristics and abilities that are most relevant in assessing DIQ in context. A survey or case study approach would be best for this.

An important contribution would be to empirically evaluate how decision makers select which among the many competing data quality dimensions are most important. This would involve development of hypotheses along with development of an experiment to test them. Decision makers must make DIQ trade-offs and if the trade-offs can be quantified then resources can be put toward improving DIQ in areas where it will have the most impact. A related area is predicting which dimensions of quality are most important in specific decision-making contexts (e.g., accuracy, security, or representational consistency). Again, these predictions can then be tested empirically. This is an area where researchers in the accounting field can contribute with their innate appreciation of the nuances of decision-making.

In summary, accounting is focused on the generation of information products by AIS, to ultimately be used in decision-making, whether internally by management or externally by investors. With the focus on fitness-for-use, new directions can be explored for improving DIQ that will better support decision-making in an increasingly competitive business world. Work that has explored the cognitive abilities of decision makers is a good start in this direction. Additionally, explicit consideration of the many dimensions of DIQ can help systems developers provide data and information that will enable decision makers to better perform their jobs.

V. CONCLUSIONS

In this paper the authors sought to motivate research addressing the following essential question: "How can we plan, design, and implement policies, procedures, systems and information products within an appropriate context, such that the quality of the data and information will most effectively support decision-making?" In order to determine if the question has been answered, it is important to look at the existing literature in the DIQ field. To facilitate this process, a detailed framework has been provided for guiding DIQ research, based on four major research strands (*people and decision-making*, *governance*, *operations*, and *technology*). The PGOT framework provides structure to create supporting DIQ research questions in addition to those provided by the authors. An essential element of the PGOT is the focus on fitness-for-use, or decision-making in context. DIQ-related articles published between the years of 1994 and 2008 were identified, and for each of the 193 articles, the predominant research focus and method were determined. Furthermore, the researchers coded all of the DIQ articles based on subtopic research questions. The coding identified research areas that are in need of further exploration, with a focus on areas where the accounting and AIS fields can substantially contribute.

DIQ impacts all areas of a business, particularly with respect to decision-making. Given that accounting information systems frequently provide the transaction level data that is used in other decision support systems, DIQ is a critical component of any AIS. The PGOT framework described in this paper can guide those who develop and maintain AIS. Additionally, a comprehensive review of the current DIQ literature shows that there is very little current research at the juncture of DIQ and AIS. Thus, the opportunity exists for both researchers and practitioners to pioneer the development of new models for ensuring that data and information are of the highest quality within the context in which they will be used by stakeholders.

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